NC STATE UNIVERSITY

ST115 Introduction to Statistics II Course Syllabus

Course Description

Students will learn the basic fundamentals of coding, which will then evolve into learning methods for discovering patterns in data through the use of effective data visualization tools, exploratory data analysis, and modeling techniques. They will gain experience working with real-world messy data, and use data tidying and data wrangling techniques to make the data suitable for exploration. An emphasis will be placed on creating reproducible and shareable work. The course will highlight the importance of and uses of version control software (GitHub) for individual and collaborative work. The course will focus on the R statistical computing language.

Learning Outcomes

Upon completion of this course, students will be able to:

- Create simple computational functions
- Develop a basic understanding on how to write mathematical equations using LaTeX.
- Develop vectorized logic, and use multi-path logic when working with vectors
- Define, install, and library open source packages
- The ability to read in data from multiple different sources
- Find and correct logical programming errors
- Use appropriate functions and tools in R/RStudio to carry out an assigned task.
- Responsibly explore, visualize, and analyze data in R/RStudio

- Create an R program or project in a reproducible and shareable manner so that it is accessible to the instructor and other students
- Implement the basic elements of version control using GitHub.
- Define tidy data in the context of this course
- Apply data wrangling and munging techniques to carry out data tidying in R/RStudio
- Fit appropriate linear models to data in R/RStudio
- Demonstrate effective communication skills
- Use classic problem-solving techniques and identify relevant code to make meaning from data

Course Materials

This course engages diverse scholarly perspectives to develop critical thinking, analysis, and debate and inclusion of a reading does not imply endorsement.

Required Textbook

All required readings in this course come from open source materials. Students will not have to purchase a textbook for this course.

Other Required Materials

Moodle, Zoom, Calculator. It is highly suggested that you have a computer that is able to download R, RStudio, and Git. Students may be asked to submit course work on Moodle/Gradescope. If online office hours are requested by the student, they will be held on Zoom or in-person in my. You will need a calculator that allows standard addition, subtraction, multiplication, division, square root and power functions. Cell phones and other electronic devices cannot be used for calculators and are forbidden on exams.

Technology Requirements

NC State University Libraries offers Technology Lending, where many devices are available to borrow for a 7-day period. Computer labs are available in various locations around campus for student use.

If your computer is unable to successfully download R, RStudio, and GitHub, you may use the NC State Posit Workbench. Posit Workbench is a secure, server-based development platform that allows you to use these technologies. The link for this can be found here.

Please contact your instructor before using the WorkBench.

Computer

A laptop computer is recommended for students taking this course. NC State's Office of Information Technology provides recommendations for your computer at NC State.

Other Devices

None

Other Student Expenses

- None

Grading

Proctored exams

All exams are closed book and closed notes and, of course, you cannot discuss the exams with other students. The exams require a proctor. The final exam is limited to 150 minutes. You must contact the instructor at least one week before the testing window for scheduled conflicts or as soon as possible for emergencies. You must take the exam with an in-person proctor within 5 business days of the original exam date. Failure to show up, or failure to give notice of absence prior to start of class on an exam day will result in a letter grade deduction (-10 points) on the exam score.

Late assignments

A 24-hour late window will be applied to homework assignments unless otherwise specified. Homework submitted during this time will receive a 10% deduction. No other assessments will be accepted late, and will receive a score of 0 unless the student provides a university approved exception prior to the deadline.

Course Assessments

This course uses these values for exams and out-of-class assignments:

Percentage of grade	Component	Details
15 %	Proficiency Test	Timed programming assignments to demonstrate basic efficiency and master the first half of the course.
5 %	Programming/Statistics Experience	The goal of the programming/statistics experience assignments is to help you engage with the statistics and data science communities outside of the classroom. You will have an experience related to the course content, and create a virtual poster summarizing your experiences.
7.5%	Attendance	Attendance is recorded and graded. This is split (2.5% each) between class and lab. 2.5% is for the completion of the lab activities.
2.5%	GitHub Teamwork Assignment	In this assignment, you will collaborate with your teammates using a shared GitHub repository. Your group will demonstrate effective version control by creating branches, making commits, submitting pull requests, reviewing each other's work, and resolving

Percentage of grade	Component	Details
		merge conflicts as needed while maintaining a clear project history.
15 %	Homework	Homework assignments will be assigned weekly / bi-weekly to help students achieve a mastery of the material.
15%	Midterm-1	A mid-semester exam to test your conceptual understanding around the fundamentals of programming.
15%	Quizzes	Quizzes will be in-person or delivered via Moodle.
25%	Final Exam	A final exam will be cumulative, and test your conceptual understanding around the relationship between data and programming. Your final exam will count toward 25% of your grade.

Attendance and Participation

Attendance is recorded and graded. For complete attendance and excused absence policies, please see http://policies.ncsu.edu/regulation/reg-02-20-03. Your lowest two attendance scores (2-days) will be dropped.

Grading Scale

This course uses this grading scale:

Low	Letter	High
97 ≤	A+	≤ 100
93 ≤	A	< 97
90 ≤	A-	< 93
87 ≤	B+	< 90
83 ≤	B	< 87
80 ≤	B-	< 83
77 ≤	C+	< 80
73 ≤	C	< 77
70 ≤	C-	< 73
67 ≤	D+	< 70
63 ≤	D	< 67
60 ≤	D-	< 63
0 ≤	F	< 60

University Policies

Students are responsible for reviewing the NC State University Policies, Rules, and Regulations (PRRs), which pertain to their course rights and responsibilities, including, but not limited to:

- POL 04.25.05 Equal Opportunity and Non-Discrimination Policy Statement
  - REG 02.20.01 Academic Accommodations for Students with Disabilities
- POL 11.35.01 Student Conduct
  - REG 11.35.05 Code of Student Conduct
  - REG 11.35.02 Student Discipline Procedures
- REG 02.20.03 Attendance Regulations
- REG 02.50.03 Grades and Grade Point Average
- REG 02.20.15 Credit-Only Courses
- REG 02.20.04 Audits
- REG 08.00.11 Online Course Material Host Requirements